

RESERVE BANK OF INDIA

Master Direction on Automated Credit Assessment and Risk-Based Lending A Template for Regulated Entities Operating AI-Driven Decisioning Systems

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This document is a **template Master Direction**. Regulated Entities (REs) — Scheduled Commercial Banks, NBFCs, and Housing Finance Companies — must review every section marked **[TEMPLATE VALUE]** and replace the example figures with institution-specific parameters approved by their Credit Risk Committee and Board. The completed document becomes the authoritative policy source ingested into the RE's AI decisioning knowledge base (RAG store).

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1. Definitions and Scope

1.1 Purpose

This Master Direction prescribes the minimum standards that Regulated Entities (REs) must embed into automated loan origination and credit decisioning systems. It covers the complete decisioning pipeline: data ingestion, individual risk signal analysis, risk aggregation, final lending decision, and adverse action communication.

1.2 Key Terms

Term	Definition
Regulated Entity (RE)	Any Scheduled Commercial Bank, NBFC, or Housing Finance Company using an automated AI/ML system for credit decisioning.
Automated Decisioning System (ADS)	Software pipeline that accepts bureau data and applicant inputs and produces a loan approval, counter-offer, or decline.
Credit Bureau Payload	The full structured JSON or XML response received from a Credit Information Company (CIBIL, Experian, Equifax, CRIF).
Risk Signal	A discrete output produced by one analytical module within the ADS (e.g., credit score band, utilisation ratio, DTI ratio).
RAG Knowledge Base	A vector store containing chunked policy documents. The ADS queries this store at runtime to retrieve applicable thresholds.
Hard Decline	An irrevocable rejection triggered by a statutory event (e.g., active insolvency) that no downstream process may reverse.
Template Value	A placeholder numerical threshold in this document. REs MUST replace every [TEMPLATE VALUE] with their own board-approved value.

1.3 Applicability

These directions apply to all REs that deploy an ADS for retail or MSME credit products with a per-account exposure exceeding INR 50,000. REs with aggregate ADS-processed portfolio above INR 500 crore must obtain annual third-party model audits.

2. Data Privacy and PII Masking Obligations

Reference: RBI Digital Lending Guidelines (Sept 2022), IT Act 2000 S.43A, DPDP Act 2023.

2.1 Directive

REs shall ensure that Personally Identifiable Information (PII) is removed from any bureau payload before that payload is processed by any machine-learning model, language model, or rule-based scorer. The sanitisation step must be the first operation in the ADS pipeline. A boolean flag confirming sanitisation must be set and validated by every downstream module before consuming bureau data.

2.2 Fields That Must Be Masked

Data Category	Field Names	Masking Method
Consumer Identity	firstName, middleName, surname	Remove key entirely
Residential Address	streetPrefix, streetName, streetSuffix, unitType, unitId, city, state, zipCode	Remove key entirely
Employment	employer name field in tradeline	Remove key entirely
Account Numbers	accountNumber on all tradeline records	Remove key entirely
Court Records	courtName, referenceNumber in public records	Remove key entirely

2.3 Processing Rules

- (a) The original bureau payload must never be mutated in place. A deep copy must be created before masking.
- (b) The sanitised payload and the original must be stored in logically separate stores with separate access controls.
- (c) Any failure in the PII masking step must halt the pipeline and generate an alert; the application must not proceed.

IMPORTANT: Processing bureau data containing PII through an AI/ML model without prior masking constitutes a data breach under DPDP Act 2023 and attracts penalties up to INR 250 crore.

3. Credit Information Bureau Usage and Score Band Classification

Reference: CIC Regulation Act 2005, RBI CIC Master Direction 2023.

3.1 Directive

REs shall classify every credit applicant into a documented score band derived from a bureau-sourced credit score. Each score band shall carry a defined maximum base lending limit, a risk flag (LOW / MODERATE / HIGH), and a fixed weight in the composite risk score. The ADS must log the score band assigned, the score used, and the policy source (RAG-retrieved or hardcoded fallback) for every application.

3.2 Score Band Thresholds

REs must define at least four score bands. Replace all [TEMPLATE VALUE] entries with board-approved figures before ingesting into the RAG store.

Band Name	Score Range [TEMPLATE VALUE]	Max Base Limit [TEMPLATE VALUE]	Risk Flag	Aggregation Weight [TEMPLATE VALUE]
PRIME	750 to 900	INR 75,00,000	LOW	0.25
NEAR_PRIME	700 to 749	INR 50,00,000	MODERATE	0.25
FAIR	650 to 699	INR 35,00,000	MODERATE	0.25
SUBPRIME	Below 650	INR 20,00,000	HIGH	0.25

[TEMPLATE VALUE] Score Band Thresholds

PRIME: score greater than or equal to 750 — base_limit_band = INR 75,00,000 — score_risk_flag = LOW — score_weight = 0.25. NEAR_PRIME: score 700 to 749 — base_limit_band = INR 50,00,000 — score_risk_flag = MODERATE — score_weight = 0.25. FAIR: score 650 to 699 — base_limit_band = INR 35,00,000 — score_risk_flag = MODERATE — score_weight = 0.25. SUBPRIME: score below 650 — base_limit_band = INR 20,00,000 — score_risk_flag = HIGH — score_weight = 0.25.

3.3 Additional Requirements

- (a) The bureau model used must be disclosed to the applicant on request.
- (b) Score extracted from primary risk model field: riskModel[0].score or equivalent.
- (c) When RAG retrieval returns policy guidance, the ADS must prefer the RAG-sourced threshold over hardcoded defaults and set llm_response_type = RAG.

4. Treatment of Public Records and Bankruptcy Events

Reference: Insolvency and Bankruptcy Code 2016, SARFAESI Act 2002, RBI Prudential Framework for Stressed Assets 2019.

4.1 Directive

REs shall analyse all public record events in the bureau payload — bankruptcies, court judgments, liens, written-off accounts — and classify them into a severity tier. Each severity tier carries an adjustment factor that reduces the base lending limit. Severe events must trigger a Hard Decline that no downstream process may override.

4.2 Severity Classification and Adjustment Factors

Severity	Trigger Conditions [TEMPLATE VALUE]	Limit Adjustment [TEMPLATE VALUE]	Hard Decline	Statutory Basis
NONE	No public records on file	1.00 (no reduction)	No	Not applicable
LOW	Non-bankruptcy records only: liens, minor court judgments without default finding	0.90	No	RBI FPC 2019
MODERATE	Bankruptcy discharged 5 or more years ago OR single satisfied judgment	0.75	No	IBC 2016 Sec 178
SEVERE	Bankruptcy discharged less than 5 years ago OR multiple unsatisfied judgments OR active insolvency proceedings OR SARFAESI listing	0.50	YES	IBC 2016 Sec 14, SARFAESI

[TEMPLATE VALUE] Public Record Severity Rules and Adjustment Factors

NONE: no public records — public_record_adjustment_factor = 1.00, hard_decline_flag = False. LOW: non-bankruptcy adverse records — public_record_adjustment_factor = 0.90, hard_decline_flag = False. MODERATE: bankruptcy discharged 5 or more years ago — public_record_adjustment_factor = 0.75, hard_decline_flag = False. SEVERE: bankruptcy within 5 years OR multiple judgments OR active insolvency — public_record_adjustment_factor = 0.50, hard_decline_flag = True. Hard decline rules: bankruptcy filed within 24 months triggers hard_decline_flag = True. Active insolvency proceedings under IBC Section 14 triggers hard_decline_flag = True. Wilful defaulter classification triggers hard_decline_flag = True.

4.3 Hard Decline Conditions (Non-Negotiable)

- (a) Bankruptcy filed within the preceding 24 months.
- (b) Active insolvency proceedings in progress under IBC 2016, Section 14.
- (c) Applicant appears on RBI Wilful Defaulter or Fraud list.
- (d) Severity classification = SEVERE as defined in clause 4.2.

IMPORTANT: Hard Decline flags are statutory obligations. Any system design that allows a Hard Decline to be overridden by scoring, business rules, or human intervention contravenes IBC 2016 and RBI norms.

5. Credit Utilisation Assessment Standards

Reference: RBI Internal Rating-Based (IRB) Guidance, Basel III Capital Framework.

5.1 Directive

REs shall compute the revolving credit utilisation ratio for every applicant using only revolving credit accounts. Instalment loans must be excluded. The ratio must be mapped to a risk level and an adjustment factor that reduces the base lending limit proportionally.

Ratio Range [TEMPLATE VALUE]	Risk Level	Adjustment Factor [TEMPLATE VALUE]	Interpretation
0% to 30%	EXCELLENT	0.95 to 1.00	Low revolving burden; strong cash-flow management
31% to 50%	GOOD	0.80 to 0.95	Moderate utilisation; acceptable
51% to 75%	MODERATE	0.65 to 0.80	Elevated; approaching stress territory
Above 75%	CRITICAL	0.50 to 0.65	Near-maxed revolving lines; high financial stress signal

[TEMPLATE VALUE] Revolving Utilisation Thresholds and Adjustment Factors

EXCELLENT: utilisation_ratio 0.00 to 0.30 — utilisation_adjustment_factor = 0.975. GOOD: utilisation_ratio 0.31 to 0.50 — utilisation_adjustment_factor = 0.875. MODERATE: utilisation_ratio 0.51 to 0.75 — utilisation_adjustment_factor = 0.725. CRITICAL: utilisation_ratio above 0.75 — utilisation_adjustment_factor = 0.575. Formula: utilisation_ratio = total_revolving_balance divided by total_revolving_credit_limit. Only accounts with revolvingOrInstallment = R are included.

- (a) Include only revolving accounts (revolvingOrInstallment = R).
- (b) The adjustment factor multiplies the credit-score-derived base lending limit.

6. Debt Exposure and Leverage Norms

Reference: RBI Financial Stability Report, Household Debt Guidelines.

6.1 Directive

REs shall assess total outstanding indebtedness across all open tradelines before extending additional credit. Only open accounts (openOrClosed = O) shall be included. Monthly obligations from this node feed the income analysis node's DTI calculation.

Exposure Risk	Debt Multiple [TEMPLATE VALUE]	Monthly Obligation [TEMPLATE VALUE]	Lending Impact
LOW	Below 2 times gross annual income	Below 15% of gross monthly income	Full lending capacity available
MODERATE	2 to 4 times gross annual income	15% to 30% of gross monthly income	Moderate capacity reduction
HIGH	4 to 6 times gross annual income	30% to 45% of gross monthly income	Significant capacity reduction
EXTREME	Above 6 times gross annual income	Above 45% of gross monthly income	Counter-offer with reduced principal required

[TEMPLATE VALUE] Debt Exposure Risk Bands

LOW: total_existing_debt below 2 times gross annual income — exposure_risk = LOW. MODERATE: total debt 2 to 4 times annual income — exposure_risk = MODERATE. HIGH: total debt 4 to 6 times annual income — exposure_risk = HIGH. EXTREME: total debt above 6 times annual income — exposure_risk = EXTREME. total_existing_debt = sum of outstanding balances on all open tradelines (openOrClosed = O). monthly_obligation_estimate = sum of monthlyPaymentAmount across open accounts.

7. Credit Inquiry Velocity Governance

Reference: RBI CIC Regulations 2006, KYC/AML Master Direction 2016.

7.1 Directive

REs shall monitor the number of hard credit inquiries made against an applicant over a rolling 12-month window. High inquiry velocity is a recognised indicator of financial distress and potential fraud. A penalty factor derived from inquiry velocity must be applied multiplicatively to the approved lending limit.

Inquiry Count (12 months) [TEMPLATE VALUE]	Velocity Risk	Penalty Factor [TEMPLATE VALUE]	Required Action
0 or 1	LOW	1.00 (no penalty)	Normal processing
2 to 3	MODERATE	0.90	Document inquiry sources; note in file
4 to 6	HIGH	0.70	Enhanced due diligence; flag for senior review
7 or more	CRITICAL	0.50	Fraud screening mandatory; referral to Risk team

[TEMPLATE VALUE] Inquiry Velocity Thresholds and Penalty Factors

LOW: 0 to 1 hard inquiries in last 12 months — inquiry_penalty_factor = 1.00. MODERATE: 2 to 3 inquiries — inquiry_penalty_factor = 0.90. HIGH: 4 to 6 inquiries — inquiry_penalty_factor = 0.70. CRITICAL: 7 or more inquiries — inquiry_penalty_factor = 0.50. Only hard inquiries (credit-seeking applications) count; soft inquiries are excluded. Penalty factor applied multiplicatively to lending capacity after other adjustment factors.

8. Payment History and NPA Classification

Reference: RBI IRACP Master Circular DBR.BP.BC.No.1/21.04.048.

8.1 Directive

REs shall evaluate the full payment history across all tradelines. The evaluation must count delinquency events (30 or more Days Past Due), identify chargeoff history, and produce a behaviour score on a 0 to 100 scale. The behaviour score feeds directly into the composite risk aggregation.

Behaviour Band	30+ DPD Count [TEMPLATE VALUE]	Chargeoff History [TEMPLATE VALUE]	Behaviour Score [TEMPLATE VALUE]	IRACP Asset Class
EXCELLENT	0	None	90 to 100	Standard
GOOD	1 to 2	None	75 to 89	Standard / Special Mention
FAIR	3 to 5	Older than 2 years	50 to 74	Sub-Standard
POOR	6 or more	Within past 2 years	25 to 49	Doubtful
UNACCEPTABLE	Multiple events	Active collections	5 to 24	Loss

[TEMPLATE VALUE] Payment Behaviour Scoring Bands

EXCELLENT: 0 delinquencies, no chargeoffs — behaviour_score range 90 to 100 — aggregation sub-score = 90.
GOOD: 1 to 2 delinquencies, no chargeoffs — behaviour_score range 75 to 89 — aggregation sub-score = 80. FAIR: 3 to 5 delinquencies OR chargeoff older than 2 years — behaviour_score range 50 to 74 — aggregation sub-score = 65.
POOR: 6 or more delinquencies OR chargeoff within past 2 years — behaviour_score range 25 to 49 — aggregation sub-score = 30. UNACCEPTABLE: multiple chargeoffs or active collections — behaviour_score range 5 to 24 — aggregation sub-score = 5. Delinquency = account 30 or more days past due. chargeoff_history = True if any tradeline has been written off.

9. Income Verification and Debt-to-Income Limits

Reference: RBI Digital Lending Guidelines Sept 2022, Fair Practices Code 2016.

9.1 Directive

REs must assess every applicant's repayment capacity through verifiable income evidence before disbursement. The Debt-to-Income (DTI) ratio is the primary affordability metric. Applications where DTI exceeds the maximum permissible limit must be declined or restructured.

9.2 DTI Calculation

DTI = (sum of all monthly obligations on open accounts plus proposed new EMI) divided by verified gross monthly income. Income must be sourced from bank statements, ITR, or authenticated salary slips. If income cannot be verified, DTI must be treated as UNACCEPTABLE.

DTI Range [TEMPLATE VALUE]	Income Risk	Affordability Flag	Required Policy Action
0% to 30%	LOW	True	Approve at full requested amount if capacity permits
31% to 40%	MODERATE	True	Approve with income documentation on file
41% to 50%	HIGH	True	Approve with enhanced post-disbursement monitoring
Above 50%	UNACCEPTABLE	False	Immediate decline or counter-offer with reduced principal

[TEMPLATE VALUE] DTI Thresholds and Affordability Rules

DTI formula: $(\text{total_monthly_obligations} + \text{proposed_new_EMI}) / \text{verified_gross_monthly_income}$. LOW: DTI 0 to 0.30 — income_risk = LOW, affordability_flag = True. MODERATE: DTI 0.31 to 0.40 — income_risk = MODERATE, affordability_flag = True. HIGH: DTI 0.41 to 0.50 — income_risk = HIGH, affordability_flag = True. UNACCEPTABLE: DTI above 0.50 — income_risk = UNACCEPTABLE, affordability_flag = False, triggers DECLINE. Missing income rule: if monthly_income is None or zero then income_missing_flag = True and DTI is classified as UNACCEPTABLE. Maximum DTI for any counter-offer option = 0.40.

10. Risk Aggregation and Composite Scoring

Reference: RBI Model Risk Management Guidelines 2024 (draft), Basel III IRB Approach.

10.1 Directive

REs shall aggregate individual risk signals into a composite score on a scale of 0 to 100 using a documented, board-approved weighting scheme. The composite score determines the risk tier and, together with the Hard Decline flag, drives the final lending decision.

Risk Factor	Weight [TEMPLATE VALUE]	Source Signal	Normalisation Rule
Credit Score	0.25	bureau riskModel score	(score minus 300) divided by 5.5, clamped 0 to 100
Payment Behaviour	0.15	behaviour_score (0 to 100)	Use raw value directly; no normalisation needed
Public Records	0.15	public_record_severity	NONE = 100, LOW = 90, MODERATE = 60, SEVERE = 30
Credit Utilisation	0.15	utilisation_risk	EXCELLENT = 90, GOOD = 60, MODERATE = 65, CRITICAL = 10
Income and DTI	0.15	income_risk	LOW = 90, MODERATE = 60, HIGH = 65, UNACCEPTABLE = 5
Debt Exposure	0.10	exposure_risk	LOW = 90, MODERATE = 60, HIGH = 65, EXTREME = 5
Inquiry Velocity	0.05	velocity_risk	LOW = 90, MODERATE = 60, HIGH = 65, CRITICAL = 10

[TEMPLATE VALUE] Aggregation Formula and Risk Tier Bands

Composite formula: `aggregated_risk_score` = sum of (`normalised_sub_score` multiplied by weight) for all 7 factors. All weights must sum to 1.00. Hard-decline override: if `hard_decline_flag` is True then set `aggregated_risk_score` = 0.0. Risk tier A: `aggregated_risk_score` >= 80 — disposition APPROVE. Risk tier B: `aggregated_risk_score` 65 to 79 — disposition APPROVE with conditions. Risk tier C: `aggregated_risk_score` 50 to 64 — disposition COUNTER_OFFER likely. Risk tier D: `aggregated_risk_score` 35 to 49 — disposition COUNTER_OFFER or DECLINE. Risk tier F: `aggregated_risk_score` below 35 — disposition DECLINE.

11. Lending Decision Criteria and Interest Rate Framework

Reference: RBI Master Direction on Interest Rate on Advances 2016 (updated 2023), Fair Practices Code.

11.1 Decision Protocol

The ADS must follow a deterministic decision protocol in the sequence below. No step may be skipped. Each step and its outcome must be logged in reasoning_steps.

Step	Condition Evaluated	Outcome if Condition Met
1	Risk Tier = F OR hard_decline_flag = True	Output DECLINE immediately. Log reason.
2	Assign interest rate per tier (see 11.2)	Set interest_rate for application.
3	Calculate maximum lending capacity	max_approved = base_limit_band x public_record_adjustment_factor x utilisation_adjustment_factor
4	affordability_flag = False	Output DECLINE. Log income constraint.
5	Requested amount <= max_approved	Output APPROVE at requested amount and tenure.
6	Requested > max_approved AND max_approved > 0	Output COUNTER_OFFER.
7	max_approved = 0	Output DECLINE.
8	APPROVE path only: calculate disbursement	disbursement_amount = approved_amount multiplied by (1 minus origination_fee_rate)

[TEMPLATE VALUE] Interest Rate Schedule and Origination Fee

Tier A (PRIME) interest rate = 7.5% per annum. Tier B (NEAR_PRIME) interest rate = 10.0% per annum. Tier C (FAIR) interest rate = 13.5% per annum. Tier D (SUBPRIME) interest rate = 18.0% per annum. Tier F (DECLINE): no rate applicable. Origination fee rate = 0.02 (2%). disbursement_amount = approved_amount multiplied by 0.98.

12. Counter-Offer and Loan Restructuring Requirements

Reference: RBI Guidelines on Restructuring of Advances 2021, Fair Practices Code.

12.1 Directive

When an applicant's requested loan amount exceeds assessed lending capacity, the ADS must not issue an outright decline. It must generate a minimum of two restructured loan options that bring DTI within acceptable bounds. All options must be communicated to the applicant in writing.

Option ID	Description	Principal Reduction [TEMPLATE VALUE]	Tenure Change [TEMPLATE VALUE]	Min Tenure
OPT_REDUCED_AMOUNT	Reduce principal; keep original tenure	15% reduction (multiply by 0.85)	No change	12 months
OPT_EXTENDED_TERM	Reduce principal; extend repayment tenure	30% reduction (multiply by 0.70)	Add 12 months to original tenure	36 months

[TEMPLATE VALUE] Counter-Offer Generation Rules

$\text{max_affordable_emi} = \text{verified_monthly_income}$ multiplied by 0.35. OPT_REDUCED_AMOUNT: $\text{proposed_amount} = \text{requested_amount}$ multiplied by 0.85, $\text{proposed_tenure} = \text{requested_tenure}$. OPT_EXTENDED_TERM: $\text{proposed_amount} = \text{requested_amount}$ multiplied by 0.70, $\text{proposed_tenure} = \text{requested_tenure} + 12$ months. Origination fee of 2% applies to disbursement on all counter-offer options. counter_offer_logic must explain why original amount was not sanctioned. original_request_dti must be logged. If no viable option can be structured ($\text{max_approved} = 0$), issue DECLINE instead.

13. Audit Trail, Explainability and Reporting Obligations

Reference: RBI Responsible AI Guidelines 2024 (draft), IT Framework for Banks.

13.1 Mandatory Audit Fields

Field	Description	Retention
application_id	Unique identifier for the loan application	Permanent
correlation_id	Request traceability identifier	Permanent
node_execution_times	Duration of each pipeline node in milliseconds	90 days
model_reasoning	Natural-language explanation of each classification	7 years
llm_response_type	RAG (policy-retrieved) or FALLBACK (hardcoded)	7 years
confidence_score	Model confidence 0 to 1 per node output	7 years
reasoning_trace	Aggregator sub-scores, weights, hard-decline flag	7 years
reasoning_steps	Step-by-step decision node logic	7 years
rag_pool	Source documents and chunk IDs used in decision	7 years
decision	APPROVE, COUNTER_OFFER, or DECLINE	7 years
decline_reason	Written explanation for any adverse decision	7 years

13.2 Adverse Action Notice

REs must communicate the reason for every DECLINE or COUNTER_OFFER to the applicant in writing within 7 working days. The explanation and decline_reason fields in the ADS output fulfil this requirement when transmitted to the applicant-facing communication layer.

13.3 FALLBACK Path Monitoring

When any node activates the FALLBACK path (hardcoded defaults used instead of RAG-retrieved policy), the event must be logged. If FALLBACK decisions exceed 5% of daily application volume, Technology Risk must be escalated.

IMPORTANT: Audit records must be stored in an immutable, tamper-proof system. RBI may request access to audit logs with 48 hours notice. Non-compliance attracts penalties under IT Act 2000 and RBI enforcement powers.